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Electricity demand mapping from open-source data for low- and middle-income countries

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ABSTRACT

Spatially resolved energy systems modelling is increasingly used to provide more accurate insights into electrification planning and infrastructure development, yet spatially resolved electricity demand data is often unavailable in low- and middle-income countries (LMICs). This study presents a novel, open-source methodology to build a high-resolution electricity demand map covering the buildings and industry sectors, and applies it to Zambia as a case study. Our approach integrates publicly available GIS data, national surveys (DHS), and official statistics. For the buildings sector, machine learning is used to map residential demand and a top-down model for services; industrial demand is assessed with a separate bottom-up process model. Our bottom-up estimates are validated against national statistics, capturing 70 % of residential and 80 % of industrial demand before final scaling. The results reveal a stark geographic concentration of consumption, with the Lusaka and Copperbelt provinces alone accounting for nearly 60 % of building demand and the vast majority of industrial demand. This granular dataset can underpin the development of spatially explicit energy system models, facilitating informed decisions on grid infrastructure expansion, optimising electrification for off-grid areas, and supporting more equitable energy access in line with Sustainable Development Goals. The methodology is designed for replicability in other countries, offering a valuable tool for researchers and policymakers across other LMICs.

Acronyms

Cu	Copper
DHS	Demographic and Health Survey
EDeMOS	Electricity Demand Mapping from Open-Source data
GADM	Global Administrative Areas
GDESSA	Gridded Dataset for Electrification in Sub-Saharan Africa
GIS	Geographic Information Systems
HREA	High Resolution Electricity Access
IEA	International Energy Agency
LMICs	low- and middle-income countries
ML	machine learning
MTF	Multi-Tier Framework
NEAS	National Energy Access Survey
NTL	night-time light
OnSSET	OpeN Source Spatial Electrification Tool
rwi	relative wealth index

SDG	Sustainable Development Goal
USGS	United States Geological Survey
UNSTATS	United Nation Statistics Division
UNSD	United Nations Statistics Division
wi	wealth index

1. Introduction

Zambia's ambition to become a prosperous middle-income country, as detailed in their National Long-Term Vision 2030 [1], depends heavily on the sustainable development of its energy system. The country's power system, over 80 % reliant on hydropower [2], is highly vulnerable to climate change impacts, as demonstrated by recent and recurrent droughts [3]. This situation led to several significant episodes of load shedding, which severely impact households and the Zambian economy. This unreliability not only slows economic activity but also undermines the country's efforts to achieve universal electricity access,

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as defined by the UN sustainable development goal 7 (SDG7). This ambition is particularly challenging as only 54 % of the Zambian population have access to electricity, with stark disparity between urban (80 %) and rural (34 %) areas [4]. Addressing these challenges requires effective, spatially informed energy planning to guide investment decisions.

Increasingly, energy systems models incorporate higher spatial and temporal resolution [5,6], because such granular approaches produce significantly more accurate results than models with a coarser resolution [6–9]. For low- and middle-income countries (LMICs), understanding the geographic distribution of electricity demand and potential is crucial for determining the best electrification strategies between access to the grid, mini grid systems or standalone systems [5], as well as for identifying opportunities for clean cooking [10] and productive electricity use [11]. The development of such granular models relies heavily on high-quality and spatially resolved data, which are progressively available thanks to advances in Geographic Information Systems (GIS) [12–14]. While the resource/supply component of spatially disaggregated power system models benefit from widely available geospatial data for renewables potentials [15,16], the spatial granularity of the demand component is understudied, an issue that is especially pronounced in Zambia and other low- and middle-income countries (LMICs).

One of the main challenges in mapping electricity demand is the availability of direct energy measurements. High spatial resolution assessments often rely on smart meter systems, which provide real-time consumption levels, in particular for gas and electricity [17,18]. This data enables applications like mapping heat demand for heating infrastructure requirements [8,19–21] but they are often limited to the case of developed countries. Indeed, smart meter systems are still not widely installed in LMICs like Zambia [22], though there are some initiatives that support their development [23,24]. Thus, studies usually harness information provided by utility or pre-paid meters [25] or spatially coarse national surveys [26]. In the absence of direct measurement, satellite based night-time light (NTL) data offers a helpful alternative. NTL data has been successfully used to estimate electricity consumption at the national or provincial level [27–29], derive electricity access tiers for sub-Saharan African countries [30] and assess other indicators such as GDP [31], wealth index [32] and energy poverty [33]. Yet, validation of spatially resolved electricity consumption using NTL at a fine spatial scale in the context of Sub-Saharan Africa remains a significant limitation [31,34].

Given these data constraints, various simpler methods are employed to estimate local demand. Modelling tools like OnSSET¹ [12] use population distribution layers as a proxy for energy needs in un-electrified settlements at a local level, a methodology that has limitations as it overlooks socio-economic variations. Energy demand assessments in the buildings sector are usually performed at the household level and these household averages are used directly for projections, but they can also be disaggregated by income deciles [35,36] or for different consumer types [37]. Machine learning (ML) algorithms offer a promising avenue to assess electricity demand at a high spatial resolution. However, these techniques often require very detailed input data coming from utility meters to train the model [38,39], which is difficult to access on a sufficient scale, and especially in an LMICs context. Among these studies, Ratledge et al. [40] use ML on satellite imagery to assess electricity access impact on local livelihoods. Lee et al. [41] developed probabilistic electricity demand forecasts and applied them at the national level, but results are only available for a few countries via Open Energy Maps² (Uganda, Ghana and Senegal).

Beyond data and methodological questions, existing research is often

limited in scope and does not provide a comprehensive picture of national electricity demand. Many spatially disaggregated assessments focus on specific sub-sectors of energy demand, usually on cooling or heating demand for households [42–44]. Beyond spatial resolution, studies have also assessed the temporal resolution of the electricity demand for heating and cooling [45,46], but for the latter [46], their model calibration relies on smart meter data primarily from developed countries, potentially overlooking specificities of LMICs. Moreover, for an economy like Zambia, the omission of the industrial sector is a critical oversight, as mining is a major driver of electricity demand. While research exists on the energy consumption of industry, they operate at a relatively aggregated level [47–49]. Research specifically including mining in LMICs rarely produces a comprehensive assessment of spatial energy demand [50–52].

Finally, identifying potential demand growth (and latent demand) is crucial, especially in the sub-Saharan Africa context, as lack of current demand is often the result of lack of energy access. This is crucial for Zambia, where the grid infrastructure needs to be designed foreseeing future demand growth. Thus, some studies also focus on assessing the latent demand in areas lacking energy access [53–56]. Some of these models [55,56] rely on very detailed input data in order to model different consumer types, which they create by synthesizing multiple sources such as household surveys, and other geospatial datasets, before combining them to assess total electricity demand at high spatial resolution.

Many of these previous studies are focused on countries from the global North and are not specific to the LMICs context. Also, they often cover only some energy vectors, and do not assess the total electricity consumption. Some rely heavily on smart meter data or disaggregated input data assumptions, limiting their replicability in other contexts. This work aims to address these limitations by presenting a new open-source methodological approach for high-resolution electricity demand mapping, and applying it in a case study of Zambia. More specifically, the approach integrates open datasets such as building footprints, population distribution, night-time light data, mining locations and publicly available survey instruments and statistical reports to estimate electricity needs at local level. Our method adheres to open-source principles, allowing transparency for easy replication and adaptation across various African countries. Thus, this contribution provides a valuable tool for researchers and policymakers, including those seeking to develop more granular and accurate energy system models for improved electrification planning in LMICs in general.

2. Method

Our method aims to assess the total electricity consumption for residential, services and industry at a high spatial resolution. The analysis is conducted on a hexagonal grid, which provides the flexibility to modify its granularity based on data availability and needs of the application. The assessment of each subsector relies on combining highly spatially resolved data with data gathered at a national level (Supplementary Material 1). The year of analysis is 2019 but can be adjusted depending on the date of the datasets.

The initial step is to map the country with hexagons (Fig. 1). The geographic signature of a country and its administrative divisions is mapped using the Global Administrative Areas (GADM) database³. The country is then divided into hexagons using H3 resolution⁴ 7, which corresponds to hexagonal cells of approximately 5.2 km² [57]. Each hexagon cell has its own identifier (ID) number, which means that the coordinate reference system (CRS) projection does not affect the hexagon perimeter.

¹ Open Source Spatial Electrification Tool (OnSSET) (<https://github.com/OnSSET/onsset>)

² <https://www.openenergymaps.org/>

³ <https://gadm.org/data.html>

⁴ <https://h3geo.org/docs/core-library/restable/>

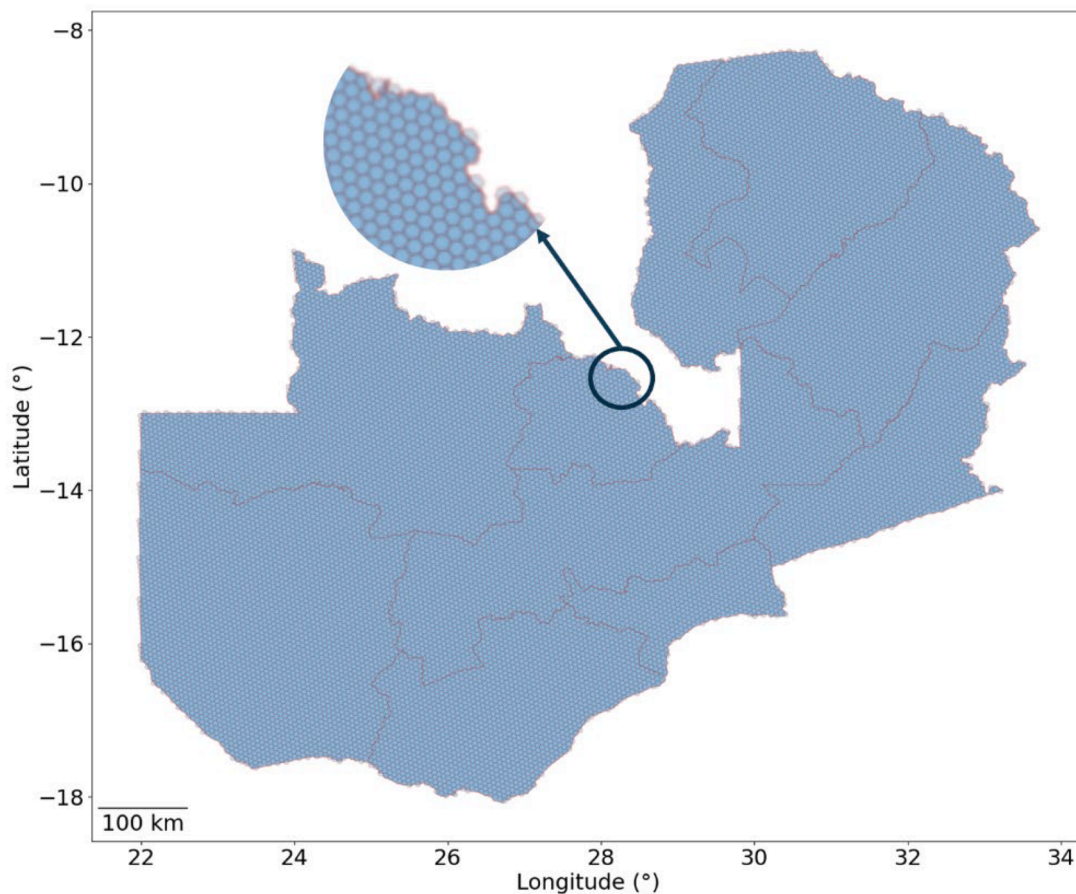


Fig. 1. Zambia's administrative map divided into H3 hexagons using granularity level equal to 7, which yields around 125,000 hexagon cells for the entire country, each approximately 5.2 km² in area (<https://h3geo.org/docs/core-library/restable/>).

2.1. Residential methodology: a bottom-up approach

The residential electricity consumption is estimated using a two-step, bottom-up approach (Fig. 2). First, the number of electrified households in each hexagonal cell is calculated. Second, the average annual electricity consumption for those households is assessed based on their wealth. The final consumption is the product of these two values. The two steps are described in more detail below and include the following datasets:

- Number of households** in each administrative division (province and/or district) derived from census data, available via the Zambian Statistics Agency [58].
- Number of buildings:** estimated via a raster layer provided by WorldPop [59] with a resolution of ~100m. Note that the number of buildings is informed from building footprints layer provided by [60]. Buildings can either be residential or services.
- Urbanisation status:** delineation between urban and rural areas as provided by WorldPop [59].
- Electrification rate:** the probability of an area having electricity access as provided by the High Resolution Electricity Access (HREA) dataset [61] with a grid resolution of 15 arcsec (~400m).

Note that (b), (c), and (d) are available for all Sub-Saharan Africa, while (a) can be typically obtained from the statistical office in each country. If this data is not available in the country, the number of

households in rural and urban areas could alternatively be obtained via the UN Population Division⁵. The urban/rural split in each subregion would then not be known, but the total number of households can still be assessed for each cell.

2.1.1. Step 1: Assessing the number of households with electricity access

The first step aims to estimate the number of households with electricity access in each hexagon cell, based on buildings counts and number of households in urban and rural areas. While the national census provides these household counts at both national and provincial levels, the provincial data in Zambia is used for greater accuracy. However, official statistics do not provide the number of households in urban and rural areas within each province ($HH_{u,p}$, $HH_{r,p}$). Therefore, these values were first estimated by applying the national urban-to-rural household size ratio to disaggregate the total household count for each province. The full methodology for this step is detailed in Supplementary Material 2.

2.1.1.1. Disaggregating households to the cell level. Once the provincial households totals for urban and rural areas are determined, they are disaggregated to the hexagon level. First, each hexagon is classified as either urban or rural based on the median value of the underlying WorldPop urbanization data (c). Then, the number of households in each hexagon (HH_{hex}) is estimated by distributing the province's total household count proportionally, based on the hexagon's share of the province's total buildings from WorldPop data (b). The calculation is

⁵ <https://population.un.org/wpp/Download/Standard/Population/>

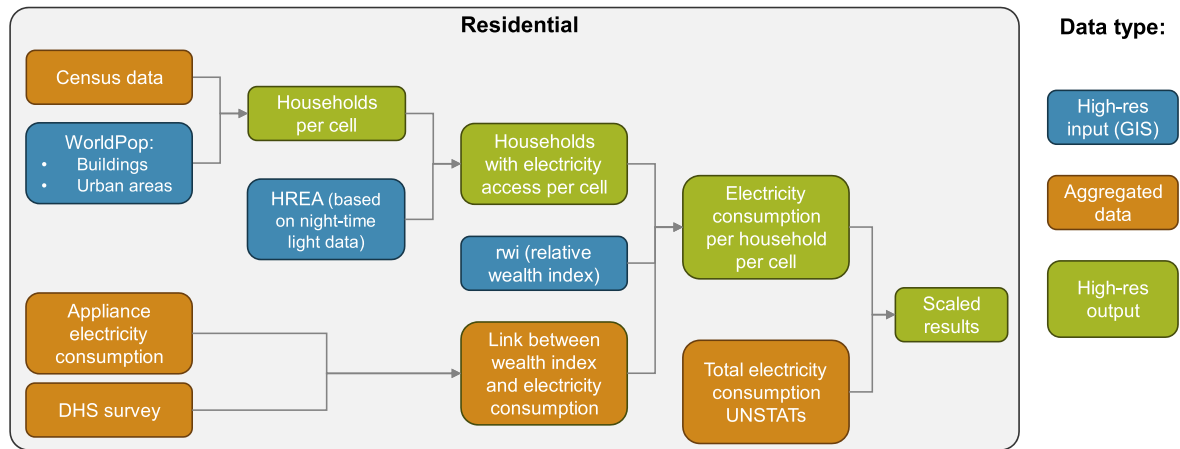


Fig. 2. Schematic representation of the methodological approach for estimating electricity demand in the residential sector leveraging GIS and non-GIS data. DHS: Demographic Health Survey UNSTATs: United Nation Statistics Division HREA: High Resolution Electricity Access rwi: relative wealth index High-res output refer to high-resolution output.

performed as follows:

$$HH_{hex} = HH_{i,p} * B_{hex} / TotalB_{i,p} \quad (1)$$

with:

- $HH_{i,p}$: total number of households in province p for area type i (urban or rural)
- B_{hex} : total number of buildings in the hexagon
- $TotalB_{i,p}$: total number of buildings in province p for area type i

2.1.1.2. Applying electricity access probability. Finally, to determine the number of households with electricity access ($HH_{access_{hex}}$), this household count is multiplied by the cell-specific electricity access probability (p_A) from data source (d), as shown in Eq. (2).

$$HH_{access_{hex}} = HH_{hex} * p_A \quad (2)$$

2.1.2. Step 2: Assess the electricity consumption per household

The following datasets are used:

- e. **Demographic and Health Survey (DHS survey)** 2018 [62]
- f. **Relative wealth index** [63,64] noted rwi hereafter

2.1.2.1. Analysis of the DHS dataset. The Demographic and Health Survey (DHS) [65] is used to estimate electricity consumptions of surveyed households. This data will be afterwards used to derive a link between the electricity consumption and the wealth index of the households in our hexagons.

For Zambia, the 2018-2019 survey is used [62]. The DHS survey gathers data from a representative sample of households across the country, with additional breakdowns at the regional level and by urban and rural areas. Along with the wealth index (noted wi thereafter), the survey also provides detailed information on household characteristics, including data on household possessions:

- Electricity access
- Radio
- Mobile telephone
- Television
- Refrigerator
- Computer
- Washing machine
- Air conditioner

– Microwave

To assess electricity consumption, each household is first assigned to an electricity access tier based on the appliances it owns and if the household has electricity access. Each appliance is then associated with a level of electricity consumption determined by the type of appliance and the household's assigned tier, using adjusted values from the ESMAP report on energy access [66] and [67] (Table 1, see Supplementary Material 3 for details). For instance, households owning only televisions are associated to Tier 2 and their televisions assumed to be lower power than those in households that also own a washing machine and are thus assigned to Tier 3.

Each household's electricity consumption E_{HH} can then be computed by adding up the electricity consumption e_i for all of the appliances i from the relevant tier, multiplied by their availability in the household $a_{i,HH}$ (binary value, 0 if not owned and 1 if owned) and adjusting the total according to the number of people in the household, n_{HH} :

$$E_{HH} = \left(\frac{n_{HH}}{n_0} \right)^\alpha \sum_i a_{i,HH} * e_i \quad (3)$$

The nominal household size, n_0 , is the number of household members that is assumed for the appliance electricity consumption values in Table 1. We adopt $n_0 = 4$. The parameter α ("the elasticity") represents how electricity use for a given house and set of appliances depends on the number of people in that house. As far as the authors are aware, specific studies assessing this relationship are not available. We adopt $\alpha=0.4$, which corresponds to two people using about a third more

Table 1

Average annual electricity consumption in kWh for a set of appliances adjusted from [66,67].

Appliance/Tier	Electricity consumption [kWh/yr]				
	1	2	3	4	5
Other electricity ¹	9	256	569	1274	1598
Mobile telephone	2	6	13	18	18
Radio	2	6	3	3	3
Television	0	29	73	128	183
Microwave	0	0	37	55	91
Washing machine	0	0	73	91	91
Computer	0	0	58	73	88
Refrigerator	0	0	0	807	1445
Air conditioner	0	0	0	0	2920

¹ Other electricity usage includes small appliances such as lighting. Higher tiers encompass additional appliances not listed above, such as dishwasher or small appliances like kettles.

electricity than one person in the same house, and a family of 5 using almost half as much again.

Thus, from the raw DHS data, a new dataset linking each surveyed household with an estimated annual electricity consumption of households, their wealth index (wi), province and area type (rural or urban) is obtained.

2.1.2.2. Analysis at the cell level. The next step consists of assessing the link between the electricity consumption and the wealth index of the different households. The core of this approach is a k-nearest neighbours (k-NN) machine learning model that leverages geographic and economic similarities between DHS survey clusters and grid cells. This non-parametric approach is chosen because it avoids imposing a potentially incorrect functional form (e.g., linear or exponential) on the complex relationship between wealth, electricity access, and consumption, allowing the data to reveal the underlying patterns. The process involves three key stages: (1) preparing a training dataset by aggregating DHS survey data to the cluster level, (2) calibrating the spatial relative wealth index (rwi) and (3) applying the stratified k-NN model to assess consumption for each cell.

- (1) The relative wealth index (rwi) in our spatial grid provided by [64] (f) has been trained with values from the clusters of the DHS dataset. Thus, to create a comparable training set, the DHS household data is first aggregated to the cluster level. For each cluster, a set of features that will be used to characterise our training set is calculated:
 - **Average wealth index:** the mean wealth index of all households within the cluster.
 - **Electricity access rate:** the share of households in the cluster with access to electricity.
 - **Average electricity consumption of connected households:** the mean electricity consumption calculated only for those households within the cluster that have electricity access.
 - **Area type:** the urban or rural status of the cluster.

While some individual households in the DHS survey exhibit very high wealth index (above 3), leading to correspondingly high electricity consumption, the clusters wi values range is much closer to the one of the rwi , meaning that there must be very few of these high-consumption households in any given cell, if any.

- (2) The spatial relative wealth index (rwi) from [64] is generated at a global level, and thus values for a specific country are not normalised, whereas the DHS wealth index has a standard normal distribution at the country level [65]. To ensure the two wealth measures are directly comparable, a regional calibration of the grid rwi is performed by using a weighted quantile mapping technique as follows: for each province within the country, the distribution of the grid rwi weighted by household counts per cell is mapped to match the distribution of the average wealth index from the DHS clusters located within that same region. At the end of the process, we end up with a standard normal distribution for the rwi , ensuring that a given wealth value is on the same scale in both distributions (Fig. 3).
- (3) Finally, to assess the average electricity consumption per household with electricity access for each grid cell, a k-nearest neighbours algorithm is used, with separate models for urban and rural areas to capture their different consumption patterns. First, a two-dimensional feature space is defined using the calibrated rwi and the electricity access rate. Then, for each grid cell, the model identifies the k most similar neighbours from the entire set of DHS clusters in the feature space, where similarity is measured by the Euclidean distance in the feature space. Finally, the estimated consumption for the grid cell is calculated as the mean of the

average electricity consumption of connected households from its k nearest neighbouring clusters. Based on cross-validation and standard heuristics, the value of k is set to 11. This value was chosen to provide a balance between model flexibility and stability, avoiding the noise associated with very small k values and the over-smoothing of large k values.

2.1.3. Final assessment

The final total residential electricity consumption per hexagon cell is then obtained by multiplying the average estimated consumption per household for the cell by the number of electrified households in this cell. In applications of this method where it is necessary that the total estimated electricity usage is exactly equal to a total from another source (e.g. national or regional electricity balance), the residential electricity use can simply be scaled to compensate for the electricity uses not included in the survey and any other limitations to the approach.

2.2. Services: a top-down approach based on occupations

In contrast to the bottom-up model for residential demand, a top-down methodology for the services sector⁶ is employed that disaggregates the national electricity consumption total from UN Statistics (Fig. 4). This requires a robust spatial proxy to allocate demand. While electricity consumption in commercial buildings is ideally driven by physical characteristics like floor area, such data are rarely available and certainly not with sufficient accuracy and classification, in particular the distinction from industrial and agricultural buildings. This presents significant challenges at a national scale in a data-scarce context like Zambia.

However, the spatial distribution of service-sector employment is available and is adopted as a proxy for economic activity and, therefore, electricity demand. This choice is supported on several grounds: first, the number of employees serves as a good proxy for the intensity of economic activity; electricity use from equipment and lighting can scale directly with active workers. Second, this approach is well-suited to LMICs and the Zambian context, where the service sector consists of many small-scale enterprises where cooling is still not widely available. Finally, this choice is supported by the data availability at the province level from the DHS survey. We acknowledge that this is a simplification and, for transparency, an alternative proxy based on building counts is presented and discussed in Supplementary Material 4.2.

Based on this approach, the methodology involves two main steps: (1) estimating the number of service-sector employees in each hexagon cell, and (2) allocating the national consumption total based on this distribution.

2.2.1. Step 1: Assessing the number of employees per cell

To estimate the number of employees, cell-level population data are combined with provincial-level employment characteristics from the DHS survey. First, the population counts per cell obtained with the residential analysis are disaggregated by sex using the national gender ratio from the census [58] and the proportion of each sex aged 15-49, as reported in the DHS survey. Second, we extract from the DHS survey on characteristics of respondents (e) [62], the share of women and men working in the following services-related occupations: Professional/technical/managerial, clerical, sales and services, and skilled manual. These shares are computed separately per province p and per area type (urban or rural) and are based on the population aged 15-49 employed in the 12 months preceding the survey.

Finally, for each cell, the number of service-sector employees (*Employees*) is estimated by multiplying its male and female populations

⁶ This sector encompasses a broad range of economic activities, including retail, finance, public administration, education, and health, and covers entities of all sizes, from Small and Medium Enterprises (SMEs) to large organizations.

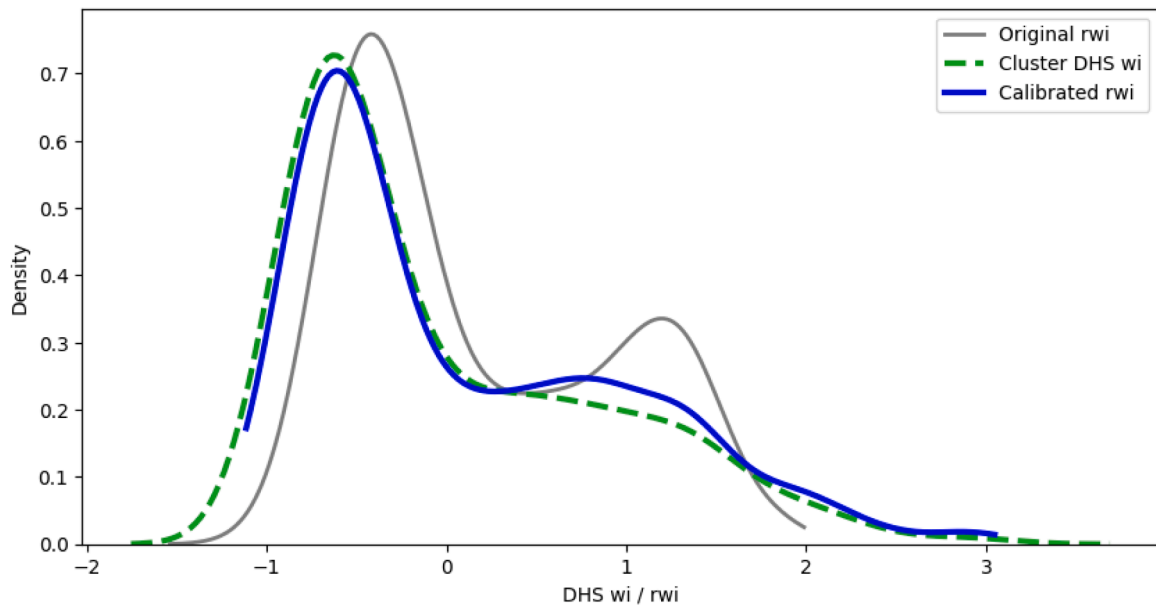


Fig. 3. Comparison of wealth index distributions The relative wealth index (Original rwi) provided by [64] is shown with the weights of the households counts, the cluster DHS wi correspond to the average wealth index in the DHS clusters, the calibrated rwi is the rwi after the quantile mapping.

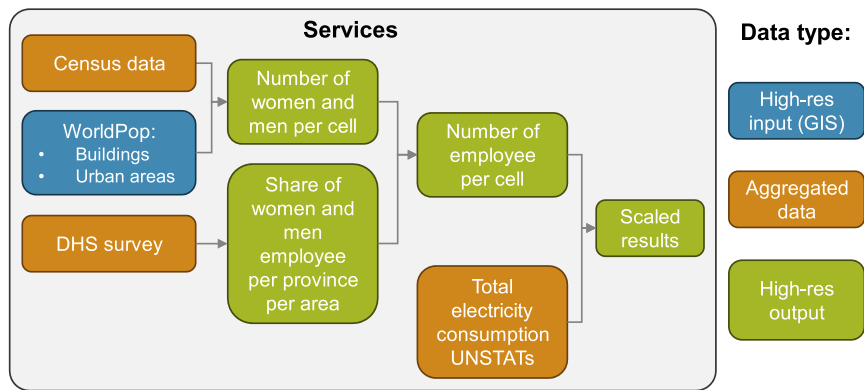


Fig. 4. Schematic representation of the methodological approach for estimating electricity demand in the services sector leveraging GIS and non-GIS data. DHS: Demographic Health Survey UNSTATs: United Nation Statistics Division.

by the corresponding provincial worker shares ($ShareWorker_{i,p}$) for its area type (urban or rural), as noted in Eq. (4):

$$Employees = Pop_{women} * ShareWorker_{i,p,women} + Pop_{men} * ShareWorker_{i,p,men} \quad (4)$$

where p is the province of the cell, and i is its area type (urban or rural).

2.2.2. Step 2: Allocating national electricity consumption

Finally, the national service sector electricity consumption (from the UN statistics) is allocated between all cells with an electrified status, proportionally to the estimated number of employees in each cell. The cell is considered electrified (binary value of 1, else 0) if the probability given by the HREA dataset is greater than a threshold value set at 90 % and 10 % for urban and rural areas, respectively. The different electrification thresholds reflect the different nature of electrification. In dense urban areas, a high probability is needed to confirm robust grid presence, whereas in rural areas, even a lower probability often indicates the presence of a mini-grid or grid extension that could power service sector activities.

2.3. Industry methodology: a bottom-up assessment

Zambian industrial electricity consumption is dominated by the mining industry, which plays a significant role in the economy. This sector contributes approximatively 15 % of GDP [68] and around 70 % of export earnings [69]. The main metal produced is copper, with cobalt as a by-product. With an annual copper production between 750 and 850 kt in recent years, Zambia ranks among the ten largest copper producers globally [70]. Although Zambia possesses considerable cobalt reserves [71], cobalt production has decreased sharply in the last decade from around 5 kt mine production in 2014 [72] to 0.2 kt in 2021 [73]. Other metals are produced, such as gold, manganese and nickel with average outputs from 2018 to 2022 of respectively 4 t, 96 kt and 3 kt (representing 0.1 %, 0.5 % and 0.1 % of global production) [74–76].

Given that the mining and non-ferrous metal subsectors account for 93 % of industrial electricity use [77], our methodology focuses on mapping this consumption with a bottom-up approach (Fig. 5). It leverages a comprehensive GIS dataset of mining sites from the USGS [78] and follows a four-step process for each key mineral, starting with copper. First, the electricity intensity (MWh/tonne) for each production process is established from a literature review. Second, the 2019 production tonnage for each individual mine site is estimated by adjusting

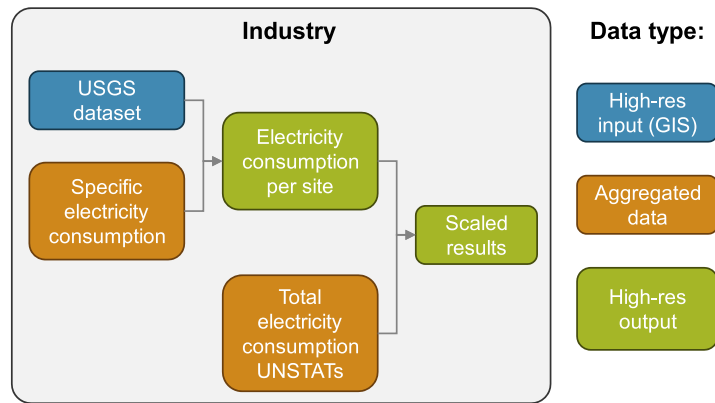


Fig. 5. Schematic representation of the methodological approach for estimating electricity demand in the industry sector leveraging GIS and non-GIS data. USGS: United States Geological Survey DHS: Demographic Health Survey UNSTATs: United Nation Statistics Division.

the 2017 USGS capacity data. Third, specific processing routes (e.g., pyrometallurgy, hydrometallurgy) are assigned to each site based on available data. Finally, the electricity demand for each site is calculate and the aggregated results are scaled to match national 2019 consumption totals from the UN statistics [77].

2.3.1. Copper production electricity demand

2.3.1.1. Copper production routes. Zambia mainly produces primary copper [79]. Two main parts of the production are ore mining and metal processing. The most energy-intensive steps in ore mining include ore extraction, transport to the processing plant and comminution (i.e., rock crushing and grinding) [80]. Subsequent processes such as filtering, drying and flotation, are less energy-intensive. Mining equipment for ore extraction and transport such as haul trucks, excavators, drills and loaders, is typically powered by diesel or other liquid fuels. Ore extraction occurs in either open-pit or underground mines. Open pit operations are generally used for mining low-grade ores (around 0.5 %), whereas underground operation is suited for higher ore grades (1–2 %) [81]. In open pit mines, energy consumption also increases with the mine's age, as haulage distances grow longer to access deeper ore bodies. In underground mines, the fuel consumption for transport is reduced, and electricity is mainly used for ventilation [81]. The comminution step, which includes crushing, grinding, and pumping, also uses electricity [80].

The metal processing route, which dictates electricity consumption, depends on the ore type (sulfide or oxide), as depicted in Fig. 6. Sulfide ore are processed via the pyrometallurgical route, while oxide ores undergo hydrometallurgical processing.

The pyrometallurgical route involves finely milling sulfide ore and concentrating it via flotation to achieve a copper concentrate with a grade of 25–35 % [82]. To produce high-purity copper, the concentrate undergoes a series of refining steps: smelting (~65 % copper content),

converting by oxidation (~98 % blister copper), fire refining and anode casting by reduction (99 % anode copper), and finally electrorefining to produce high-purity cathode copper (99.95–99.99 % copper) [81,83].

The hydrometallurgical route, in contrast, does not require such fine milling. Instead, crushed oxide ore is subjected to leaching, where copper is dissolved into a solution. Following solvent extraction to purify this solution, cathode copper is produced via electrowinning [81]. This two-stage process is often referred to as solvent extraction and electrowinning (SX-EW).

In addition to these two production routes, copper is also processed in non-ferrous metal foundries, where it is melted and cast—among other things—into copper casting alloys. Beyond foundry products, copper metals are likewise used to produce semi-finished products such as bars, tubes, and wires. In this study, however, the production of these semi-finished products is not considered because of the heterogeneous industry structure and the lack of sufficiently detailed data.

2.3.1.2. Assessing site-specific electricity consumption. The electricity consumption of each copper site is therefore determined by several factors, including mine type, mine location, ore grade, metal processing route and equipment employed (e.g. smelting, converting and flotation). Our methodology accounts for these factors through the following four steps.

First, a thorough literature review was conducted to determine the specific energy consumption for each process in the copper production chain, as summarized in Table 2. The literature typically provides energy intensities per ton of copper produced. However, the USGS dataset reports production capacity primarily in tons of processed ore or concentrate. Therefore, a critical step was to estimate the average ore and concentrate grades to standardise these units.

The average ore grade g_{avg} of Zambian mines was calculated by weighting the average mill grade $g_{mill,i}$ of each mine i reported in [85] with the quantity of its milled ore $Q_{mill,i}$, as shown in Eq. (5). When a

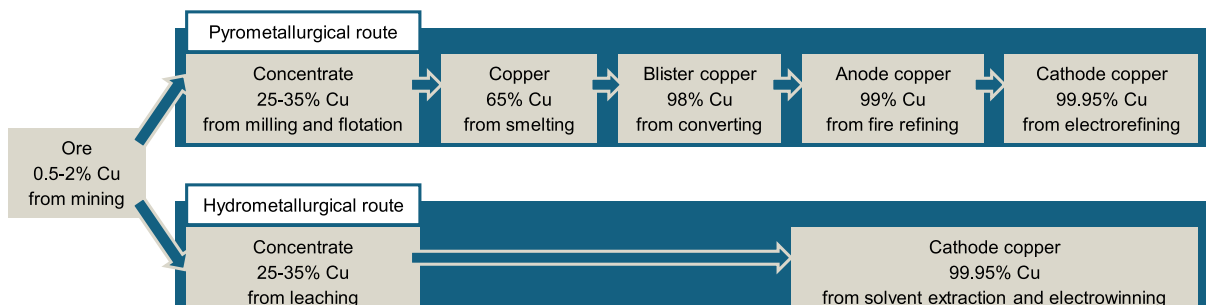


Fig. 6. Copper transformation steps and the copper content in respective processes.

Table 2

Copper production processes and their electricity intensities.

Product type	Process	Process type	Electricity intensity [MWh/t _{Ca}]	Reference
Ore and concentrate	Mining	Open pit	-	[81]
Ore and concentrate	Mining	Underground	0.60	[81]
Ore and concentrate	Milling		2.67	[80]
Metal	Smelting	Flash smelting, flash converting	2.57	[84]
Metal	Smelting	Isasmelt, Peirce Smith converting (PSC)	1.92	[84]
Metal	Smelting	Mitsubishi Process	2.36	[84]
Metal	Smelting	Noranda–Teniente	2.80	[84]
Metal	Smelting	Noranda reactor, PSC	2.48	[84]
Metal	Smelting	Bottom blowing smelting, PSC	2.57	[84]
Metal	Smelting	Reverberatory furnace	0.60	[84]
Metal	Refining	Electro-refining	0.36	[81]
Metal	Leaching, SX EW		3.47	[80]
Metal	Foundry		1.60	[83]

mine's mill grade was unavailable, it was estimated by dividing its copper production capacity with its ore milling capacity. If these values were also missing, the most recent available annual mill grade for that mine was used. Using this methodology, the weighted average ore grade for Zambian mines was estimated to be 1.03 %. Based on [82], the copper concentrate grade was assumed to be a standard 30 %.

$$g_{avg} = \frac{\sum_i g_{mill,i} \cdot Q_{mill,i}}{\sum_i Q_{mill,i}} \quad (5)$$

The second step establishes the production level for each site for the year 2019. First, the product type for each site in the USGS dataset was verified and adjusted using accompanying notes to ensure capacity was assigned the correct unit (tons of ore, concentrate or metal). This verification revealed that no sites were classified with 'concentrate' as their final product type. Next, data gaps were addressed. Negative capacity values, which indicate missing data, were set to zero. For the Kansanshi and Sentinel mines, however, these gaps were filled using 2019 production data from [86,87]. To adjust the 2017 dataset capacities with the 2019 study year, a national usage rate for each product category is calculated. The total productions - of ore and concentrate, and of metal - were scaled to match the national 'mined copper' and 'refined copper' production totals reported by the USGS for 2019 Yearbook [79], resulting in usage rates of 77 % and 31 %, respectively. Finally, since energy intensities are expressed per tonne of copper content, the scaled production volume for ore was multiplied by the estimated average ore grade (1.03 %) to yield the final copper tonnage for each site.

In the third step, specific production processes were assigned to each site based on keywords in the site name⁷. For example, a site name containing 'plant' or 'facility' was assumed to contain both smelter, refinery and foundry processes, while a name explicitly including 'smelter' indicated only that single process. Refineries were assumed to also host the foundries. To identify the production type (e.g., Isasmelt, Mitsubishi, Flash), site names were further scanned for relevant keywords. In the case that no further information can be extracted, default types were assumed, namely an underground mine for mining sites, and flash smelting for a smelting site.

In the final step, the total electricity demand for each site was calculated by multiplying the site's estimated copper tonnage (from Step 2) by the sum of the specific electricity consumption values (from Table 2) for all its assigned processes (from Step 3).

2.3.2. Cobalt, gold, nickel and manganese production electricity demand

Following a similar procedure as for copper, the electricity required

for cobalt, gold, nickel and manganese production was estimated.

The first step determines electricity intensity per tonne of the produced metal. Globally, cobalt is mainly extracted as a by-product of nickel mining (about 55 %) and copper mining (about 35 %) [88]. Cobalt ore in Zambia is produced in the same mines as copper and nickel and thus mainly shares their capacity in the USGS GIS dataset. Moreover, cobalt ore grade significantly differs across the Zambian mines [89]. Thus, only the metal production process is considered for the cobalt production. Furthermore, the USGS GIS dataset gives information on production capacities for gold, nickel and manganese in Zambia directly as a mass of metal content. Therefore, no estimation of ore or concentrate grade is necessary. Energy intensities for each metal and production process are given in Table 3.

According to the literature survey from Allen [80], the total energy consumption per ton of produced gold for mining and metal processing considerably differs for underground (37 MWh/kg) and open pit mines (103 MWh/kg), due to the typically higher ore grade in underground mines. The metal processing part in the gold production contains cyanide leaching, carbon loading, electrowinning and tailings management [80]. Energy intensity of the nickel sulfide mining operations does not differ significantly between the underground and open pit mines [80], wherefore an average value from that source was taken.

In the next steps electricity usage per site is determined. A usage rate of produced materials is calculated so that the sum of the site capacities from the USGS GIS dataset multiplied with the usage rate fits the

Table 3

Cobalt, gold, nickel and manganese production processes and their electricity intensities. Values marked by * represent total energy intensity instead of electricity intensity.

Material	Product type	Process (type)	Electricity intensity [MWh/t _{metal}]	Reference
Cobalt	Metal	Refining	3.77	[90]
Gold	Ore and concentrate	Mining (Open pit)	-	[80]
Gold	Ore and concentrate	Mining (Underground)	6.70 · 10 ³	[80]
Gold	Ore and concentrate	Milling (Open pit)	26.86 · 10 ³	[80]
Gold	Ore and concentrate	Milling (Underground)	9.67 · 10 ³	[80]
Gold	Metal	Processing (Open pit)	14.47 · 10 ³ *	[80]
Gold	Metal	Processing (Underground)	10.79 · 10 ³ *	[80]
Nickel	Ore and concentrate	Mining, Milling	5.00	[80]
Nickel	Metal	Smelting, Refining	55.56 *	[80]
Manganese	Ore and concentrate	Milling	0.01	[91]
Manganese	Metal	Refining	5.00	[91]

⁷ Additional notes partly provide the information if the output product was anode or cathode copper. However, this information does not specify the starting copper product or the production route. Thus, the note does not reveal (all) involved processes. If available, this information was therefore used for the validation of the assumptions.

national production from the USGS mineral yearbooks. Next, appropriate electricity intensity per site and material is selected. As the USGS GIS dataset contains no additional information regarding gold mine type, the average value is used. Finally, the multiplication of the production capacity, usage rate and energy intensity deliver the final electricity estimates.

2.3.3. Summarising and adjusting electricity demand

The electricity consumption of all sites are then scaled so that the combined total electricity use matches the total electricity consumption of non-ferrous and mining sector for the year 2019 based on UN statistics data [77] to ensure the accuracy and representativeness of the overall assessment.

3. Results and discussion

This section presents the high-resolution electricity consumption estimates for the residential, services, and industrial sectors. The bottom-up results are compared against national benchmarks as well as independent subnational and operator-level datasets. Finally, the implications of these findings alongside data uncertainties and model limitations are discussed.

3.1. Residential and services

3.1.1. Spatial distribution of electricity demand

The residential bottom-up approach leads to a total electricity consumption of 2,802 GWh. This accounts for around 70 % of the 4,023 GWh benchmark reported by the UN statistics for 2019 [77]. This underestimate can be expected, as our approach based on the DHS survey does not capture all electricity uses and may not fully encompass all residential activities and other uses that might be included under

residential such as small-scale productive uses or informal grid consumption. Moreover, the national statistics are derived from utility sales data based on tariff categories. Consequently, they might include consumption from small businesses, informal connections, and other non-household activities that are billed under residential tariffs but are invisible to our appliance-based model. To align our result with national accounts, a scaling factor of 1.44 is applied to our residential estimates. This approach acknowledges the inherent uncertainty in national energy data, where different sources like the Energy Regulation Board (ERB) report different totals, highlighting the challenge of establishing a single definitive benchmark (see Table 5 and Supplementary Material 6). For instance, ERB reports for the broader buildings sector a consumption of 4,477 GWh. The service sector consumption is, by design, aligned with UN statistics. Our top-down approach leads to an electricity consumption per estimated employee of 323 kWh. All subsequent results for the buildings sector (residential and services combined) are based on these scaled figures.

The primary contribution of our work is the high-resolution mapping of electricity demand. Fig. 7 illustrates the final electricity consumption for the buildings sector, revealing stark geographic inequalities (see also Supplementary Material 7). The electricity consumption is heavily concentrated in Zambia's economic hubs: the capital city of Lusaka and the Copperbelt province, which concentrates most of the mining activities. As detailed in Table 4, these two provinces alone account for nearly 60 % of the total building electricity demand (33 % in Lusaka and 26 % in Copperbelt). This pattern reflects the higher urbanisation, wealth, and industrial activity in these regions. In stark contrast, vast rural areas exhibit minimal consumption, corresponding to lower electrification rates and economic activity. As expected, the main consumption centres align closely with the national transmission grid infrastructure.

Validating spatially disaggregated energy models is especially difficult due to the scarcity of official granular data in LMICs. For this case, as

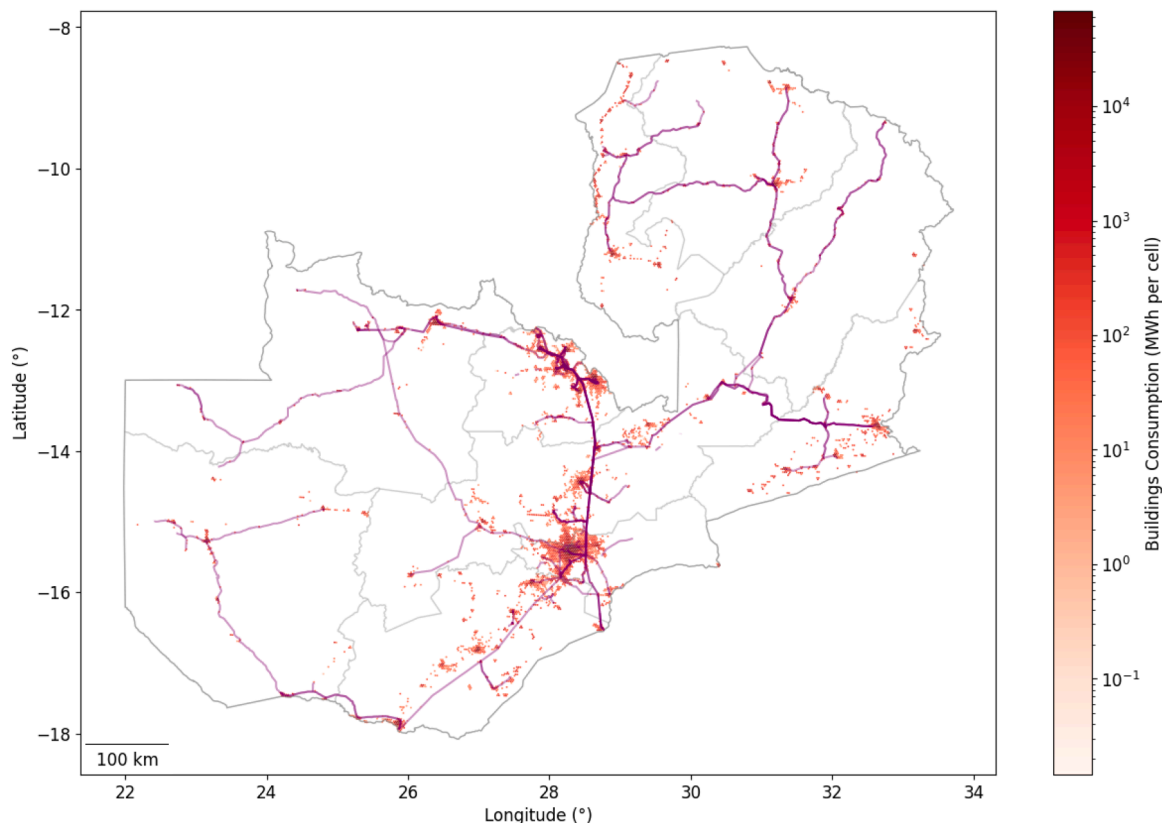


Fig. 7. Electricity consumption in the buildings sector (MWh) The red intensity corresponds to the electricity consumption in the buildings sector in kWh on a logarithmic scale. Purple lines are the Zambian transmission lines.

Table 4
Electricity consumption per region and per sector in 2019 (GWh).

Province	Residential		Services	Buildings	Share (buildings)
	before scaling	after scaling			
Central	223	321	52	373	8 %
Copperbelt	705	1,012	201	1,213	26 %
Eastern	175	252	42	294	6 %
Luapula	142	203	25	228	5 %
Lusaka	926	1,329	241	1,570	33 %
Muchinga	31	45	7	52	1 %
North-Western	128	184	37	221	5 %
Northern	106	152	25	177	4 %
Southern	269	386	65	451	10 %
Western	98	140	26	166	3 %
Zambia	2,802	4,023	722	4,744	100 %

the best available proxy, our provincial results are compared to a dataset from the Cities and Infrastructure for Growth Zambia (CIGZ) programme⁸. However, a direct comparison with this source requires acknowledging significant inconsistencies between the reported consumptions (Table 5). The CIGZ data, which only provides a breakdown between the mining sector and the rest, reports a much larger non-mining consumption total (6,862 GWh) than UN statistics (5,753 GWh).

With this context, a comparison of provincial consumption shares (Table 6) shows that our model aligns well with CIGZ data for most provinces, with differences typically within a few percentage points. The most notable discrepancy is in Lusaka, where our estimate is lower (33 % vs. 50 %). This divergence likely stems from differences in sectoral scope and urban complexity. First, the CIGZ 'Other' category aggregates buildings, agriculture and unspecified industry—sectors often located in the capital—which are excluded from our building-specific model. Second, our bottom-up approach, using building footprints, may conservatively estimate consumption in dense urban cores by not fully capturing multi-story apartments or informal connections. However, the close alignment in provinces where these distorting factors are minimized (e.g., Central) supports the robustness of our spatial allocation. These differences underscore the value of our bottom-up, sector-specific methodology in disentangling consumption patterns that are aggregated in other datasets.

3.1.2. Key drivers of consumption

3.1.2.1. Electricity access rates. A key driver of our results is the spatially explicit electrification rate, which is derived from the High Resolution Electricity Access (HREA) dataset [61]. Based on this, our model estimates a national electricity access rate of 39 % for Zambia in 2019. As shown in Fig. 8, this result is highly consistent with figures from official sources, falling within the reported range of 34 % (IEA, DHS) to 43 % (World Bank, MTF). While some sources report at the household level, and others at the population level, the difference is

Table 5
Electricity consumption per sector in 2019 (GWh).

	CIGZ (GWh)	UN stats (GWh)	ERB (GWh)
Total	13,250	12,341	12,831
Mining	6,388	6,588	6,551
Other	6,862	5,753	6,280
Buildings		4,744	4,477
<i>Other</i>		1,009	1,803

⁸ Data has been provided through personal communication.

Table 6
Electricity consumption per province in 2019 (GWh).

Province	CIGZ		This study	
	Consumption (GWh)	Share	Consumption (GWh)	Share
Central	500	7 %	373	8 %
Copperbelt	1,843	27 %	1,213	26 %
Eastern	179	3 %	294	6 %
Luapula	77	1 %	228	5 %
Lusaka	3,409	50 %	1,570	33 %
Muchinga	64	1 %	52	1 %
North-Western	135	2 %	221	5 %
Northern	92	1 %	177	4 %
Southern	492	7 %	451	10 %
Western	70	1 %	166	3 %
Zambia	6,862		4,744	

typically minor (<2 % in the DHS survey) and falls within the overall uncertainty range. This alignment provides strong initial validation for our model's national-level accuracy, despite the well-documented uncertainties surrounding the electricity access rates estimations, which have been recently highlighted by Hirmer et al. [92].

The strength of our model lies in its ability to capture sub-national variations. An urban access rate of 72 % and a rural rate of 15 % is estimated. To ground our model, the urban rate was calibrated to align with national statistics, as the raw HREA data indicated near-universal access in urban centres. Thus, our urban estimate aligns with the consensus from the MTF, World Bank, and DHS surveys (all ~75 %). Our rural access rate of 15 % is also close to the ~12 % reported by these same surveys. While this slight overestimation may lead to a marginally higher estimate of rural electricity consumption, it accurately reflects the spatial distribution of night-time light captured by the HREA methodology.

Electricity access rates can also be validated by consumption tiers (Fig. 9), corresponding to different levels of energy consumption as defined in the MTF classification [66]. Two datasets, MTF [95] and GDESSA [30] provide breakdowns by tier, although MTF's Tier 4 and 5 are grouped together as Tier 4 in GDESSA. It is important to first note that these benchmark datasets show significant discrepancies on top of their inherent uncertainties, as underlined previously, which limit their reliability for direct validation.

In urban areas, our model's tier distribution is broadly similar to the MTF, though it concentrates more households in Tier 4, while the MTF reports a higher share in the top-end Tier 5. In rural areas, the differences are more significant. While both show similar numbers of un-electrified households (Tier 0), our model allocates the majority of connected rural households to higher consumption tiers (primarily Tier 4), in stark contrast to the MTF, which places most in Tier 1.

This discrepancy highlights a key limitation of the cell-based approach and these tier-level comparisons should be interpreted with caution. The difference in rural tiers likely arises because the methodology assigns a single, average consumption profile to all electrified households within a hexagon. In reality, urban and rural areas contain significant heterogeneity, and the model's average may be skewed by smaller pockets of higher-income households, thus inflating the estimated consumption tier for the cell as a whole. Nonetheless, the analysis confirms that while the model may smooth over individual variations, it successfully captures the fundamental geographic divide between urban and rural areas in both access rates and overall consumption intensity.

3.1.2.2. Household consumption level. The model reveals a clear urban-rural divide in household electricity consumption intensity, driven by differences in wealth and appliance ownership. According to our method, the average annual consumption for electrified households in urban cells is approximately 2,000 kWh, typically ranging from 800 to 3,500 kWh in cells with a relative wealth index (*rwi*) between 0.5 and 1.5. The annual average in rural areas is lower at 1,600 kWh,

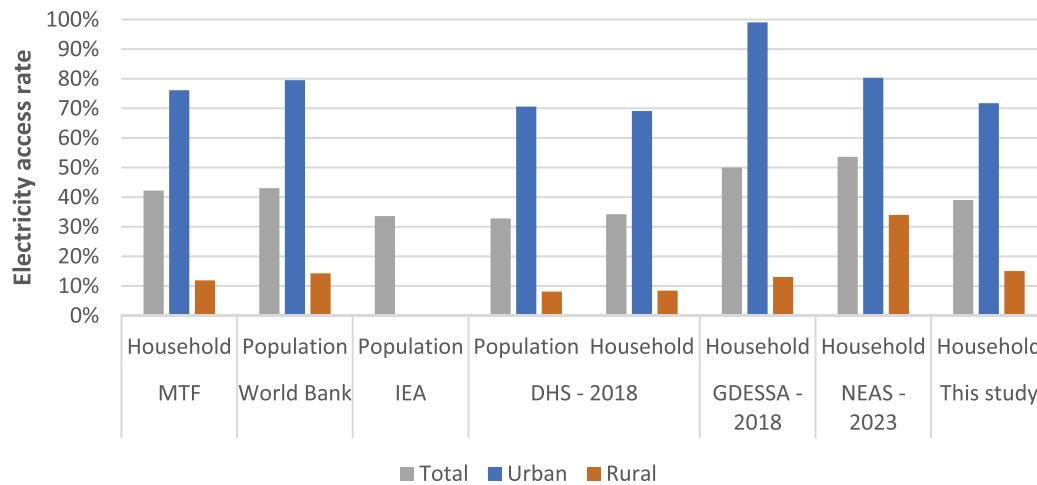


Fig. 8. Electricity access rate at the national level.

Note: If no indication, the reported year is 2019. Although the rates can be assessed at the household or population level, the differences between the two are less than 2 % percentage point in the DHS survey. This suggests that the values for both levels are likely to be close and fall within the overall uncertainty range. Source: World Bank [93], IEA [94], the Multi-Tier Framework (MTF) survey [95], the Demographic and Health Survey (DHS) [62], the Gridded Dataset for Electrification in Sub-Saharan Africa (GDESSA) [30], National Energy Access Survey (NEAS) [4]

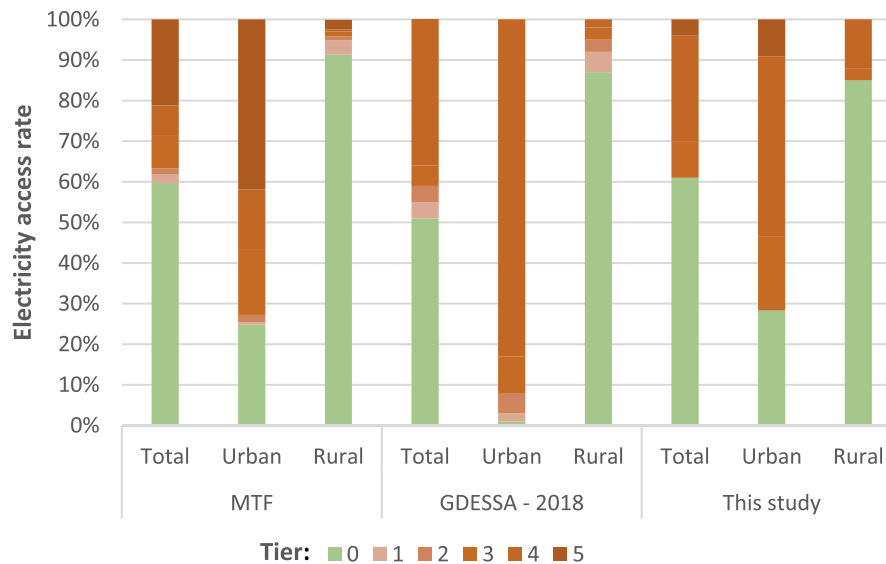


Fig. 9. Electricity access rates per tier.

Note: In the MTF data, the value for Tier 0 is higher than in the table with national access rates, as some households with electricity access do not meet Tier 1 requirement. In the GDESSA dataset, only 4 tiers are included, Tier 4 corresponds to Tier 4 and 5 from the MTF classification.

corresponding to cells with an *rwi* generally between -1.0 and 0. This difference underscores the importance of a spatially explicit approach for accurate demand assessment.

Validating these granular estimates is challenging, as publicly available data on per-household consumption in Zambia is scarce and often contradictory. Different sources - from local case studies to national reports - produce a wide range of estimates, highlighting the absence of a single, definitive benchmark. Our findings should be interpreted within this context of inherent data uncertainty.

Our urban consumption of 2,000 kWh/year is consistent with several local studies. For instance, in Kitwe, the second largest city after the capital Lusaka in Zambia located in the wealthy Copperbelt province, Mulogonti et al. [96] report an annual consumption of approximately 2,500 kWh (pre-load shedding), which aligns well with the upper end of our modelled urban range. In contrast, another survey-based estimate, the MTF [95], suggests much lower electricity consumption levels. Our

analysis of the MTF dataset leads to an urban average of around 1,300 kWh and a rural of around 900 kWh. However, these lower figures present a significant challenge: when scaled by national electrification rates, they would result in a total residential consumption far below the officially reported national statistics. This suggests that, while valuable, the MTF survey may underestimate average consumption for connected households.

Conversely, other estimates derived from utility data suggest higher averages. For instance, dividing ZESCO's total residential sales in 2019 by its reported customer base (around 900,000) yields a figure of around 4,500 kWh per customer [97]. Similarly, the Integrated Resource Plan [98] points to averages around 3,500 kWh for more recently connected customers. The difference can be explained by the methodology: utility data may be skewed by the widespread practice of shared meters, where one customer serves multiple households, a practice affecting up to 12 % of all households [4] and the potential inclusion of small commercial

enterprises under residential tariffs. This illustrates the difficulty in comparing household-level models directly with aggregate customer data.

Finally, to further validate our approach, an internal consistency check using the DHS survey itself is performed. By averaging our assessed electricity consumption of each household in the survey using their statistical sampling weight, we can create a national estimate directly from the source data. This process yields two figures: an average of 710 kWh across *all* households (including unelectrified), and an average of 2,074 kWh for *electrified* households only. Multiplying these averages by their respective total number of households yields estimated total residential consumption figures of 2,599 GWh and 2,973 GWh, respectively. Our model's final spatially distributed estimate of 2,803 GWh falls squarely between these two DHS-derived bounds. This provides strong evidence that our bottom-up spatial model accurately reflects and distributes the underlying consumption patterns captured in the national DHS survey.

3.2. Industry

Our bottom-up model for the industrial sector estimates a total electricity consumption of 5.2 TWh for 2019. This figure covers nearly 80 % of the corresponding electricity demand reported by the UN statistics [77] for the non-ferrous metal and mining subsector (Fig. 11). To align our result with national accounts, a scaling factor of 1.26 is applied to our industrial estimates. As illustrated in Fig. 10, this demand is highly concentrated, spatially allocated to specific mining and metal production sites across the country, mainly in the Copperbelt region, reflecting the industry's centralized nature. For the copper sector only, our total electricity consumption amounts to 4.5 TWh (see Fig. 11 on the left side), covering thus 87 % of the Zambian non-ferrous and mining sectors' electricity demand.

The difference between our estimate and the value reported by the UN statistics could stem from a range of factors. First, our model relies on literature-based energy intensities that may not fully capture the specific, and potentially lower, operational efficiencies of Zambia's industrial facilities. Second, a likely source of discrepancy is the relatively simplified bottom-up representation of the processes, therefore neglecting additional site-related electricity consumption like lighting, plant monitoring, water treatment etc. Furthermore, the UN energy balance only includes electricity entries for non-ferrous metals, mining and quarrying, and a negligible electricity consumption (0.1 %) for the construction industry, but it is apparent from economic data that other industries are also present in Zambia. Although there is a small portion of 6.6 % of the electricity consumption assigned to the industry consumption not elsewhere specified, it seems possible that electricity consumption in these other industries has been misattributed to non-ferrous metals and mining and quarrying.

Beyond the aggregate total, our analysis reveals a discrepancy in the allocation of consumption between the mining and the non-ferrous metal subsectors. Our model attributes roughly the same electricity shares to both, while UN statistics attributes almost all electricity consumption to the non-ferrous sector. The difference likely comes from different attribution of processes to the subsectors in the two sources used – USGS GIS dataset and UN statistics (see also Supplementary Material 8 for an allocation by process). Specifically, ore milling and concentration, an electricity-intensive process, is in this study associated with the mining sector, whereas it may be associated with the non-ferrous sector in UN statistics. This finding highlights the importance of transparent, process-based modelling to understand the true drivers of industrial energy demand for each sub-sector.

To further validate our model at the operator level, a second comparison was performed against aggregated 2021 consumption data from the IRP report [98] for the copper sector. As this data is aggregated by

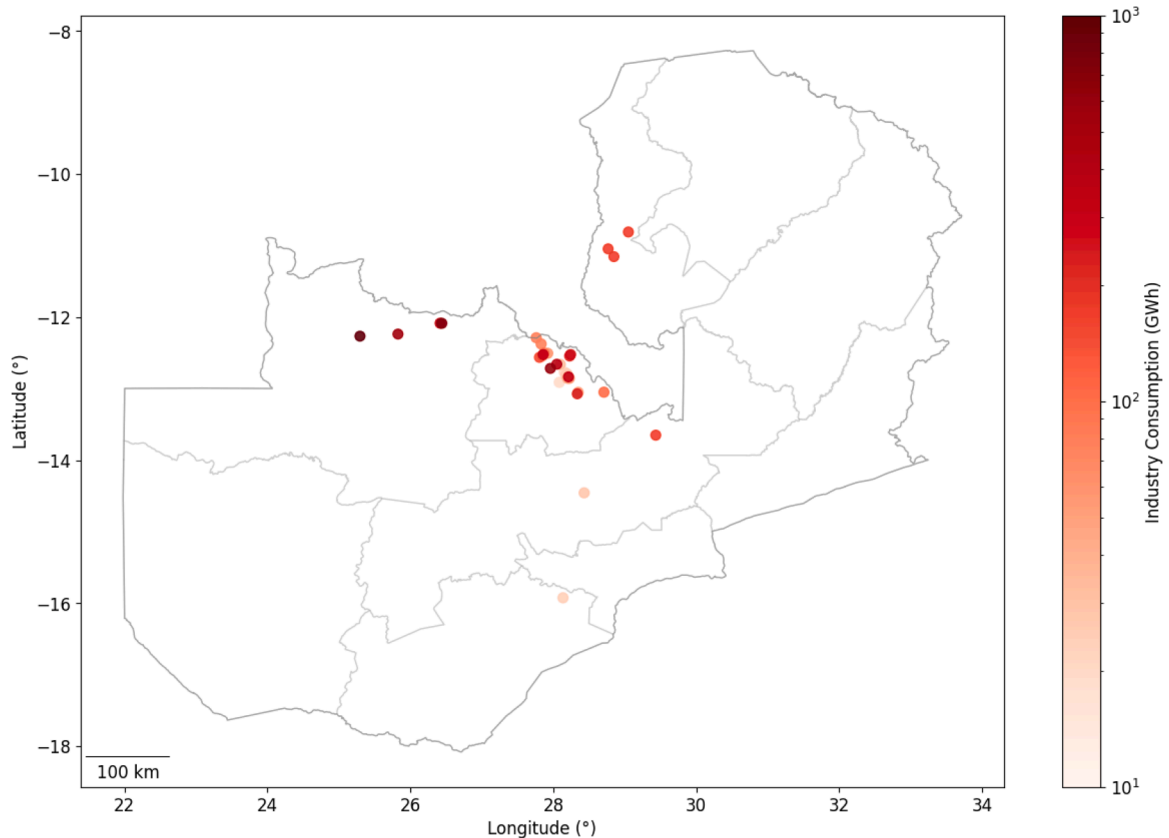
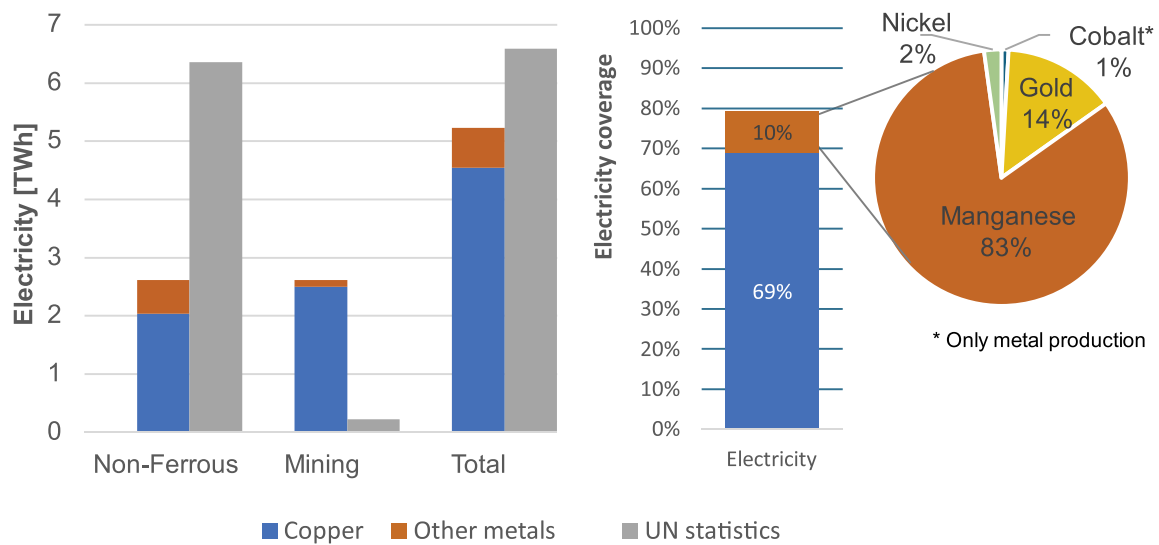


Fig. 10. Assessed industrial electricity demand per hexagon. The map shows sparsely distributed electricity demand allocated to the mining and metal production sites. Note: To enhance visibility of isolated mining sites, consumption is represented by cell centroids with increased marker size rather than standard representation.



1

Fig. 11. Total estimated mining industries electricity consumption before scaling (left) and coverage rates in comparison to the UN statistics (middle) with the shares of other metals in the coverage rate (right).

mining operator and for a different year, a direct comparison required two adjustments to our model. First, our site-level production estimates were scaled from 2019 to match the official 2021 national copper production totals reported by the USGS (see Supplementary Material 9 for detailed scaling factors). Second, our site-level electricity demand

estimates were aggregated by their respective operating companies, as identified in USGS data. After grouping, approximately 22 % of our modelled demand could not be attributed to the operators listed in the IRP report and was categorized as “Others” (see Fig. 12). Crucially, our scaled estimate covers 91 % of the electricity consumption reported for

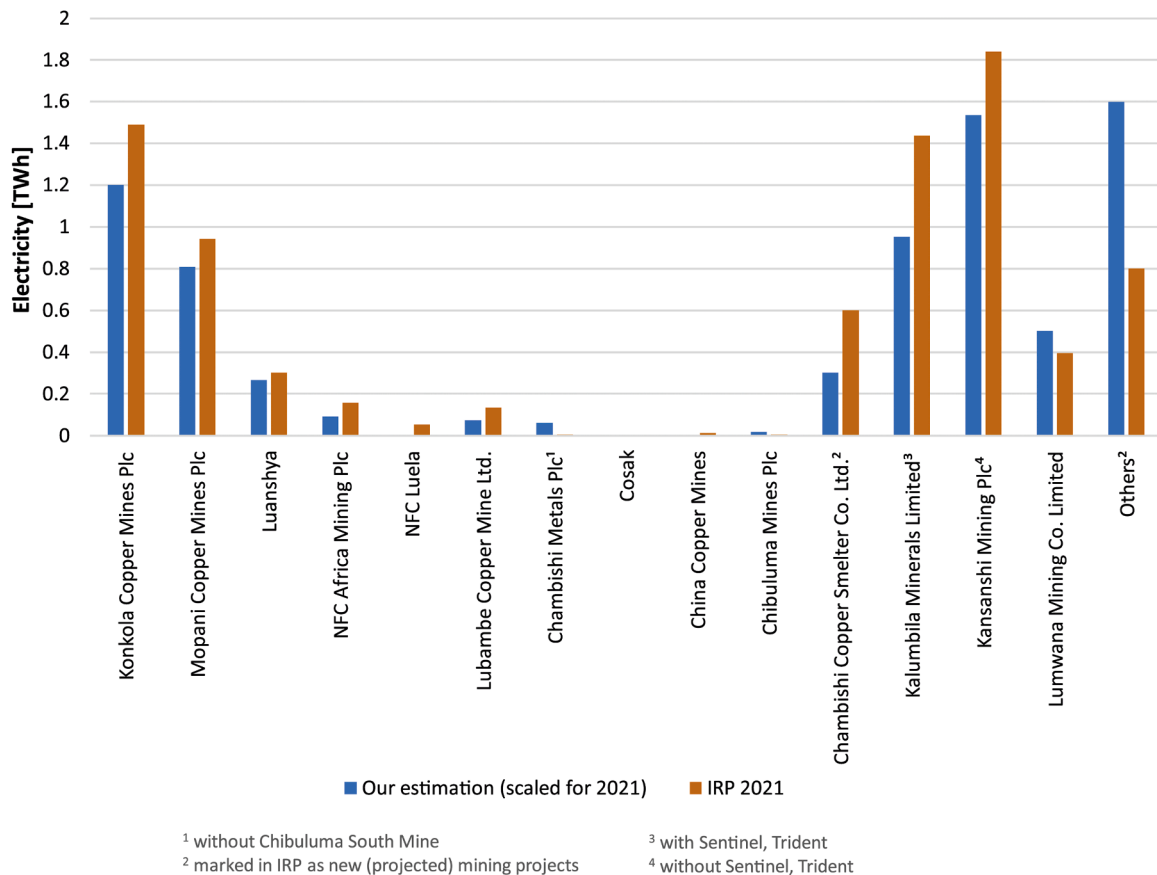


Fig. 12. Comparison of modeled and reported copper sector electricity demand by operator, 2021. Modeled demand was estimated by scaling our 2019 site-level results to match 2021 national copper production totals. Reported demand is from the Integrated Resource Plan [98]. The “Others” category represents modeled demand from sites not attributed to the major operators listed in the IRP.

mining in 2021 in the IRP. This high level of agreement provides strong, independent validation for our bottom-up model's ability to accurately capture operator-level electricity consumption.

3.3. Data uncertainties and methodological limitations

3.3.1. Data sources uncertainties

Our methodology integrates multiple open-source datasets to allow for replicability. However, each source carries inherent limitations and potential biases; it is essential to understand these before interpreting the model's scope and results.

The primary source of uncertainty in our residential access estimates is the HREA dataset. As a proxy based on Night-Time Lights (NTL), it inherits the specific limitations of NTL processing. These include the "blooming" effect, where light spillover causes over-saturation in dense urban areas, potentially leading to an overestimation of access (a bias corrected for in the urban calibration). Conversely, the model may underestimate access in rural areas due to the difficulty of detecting small, isolated off-grid systems or low-consumption connections that fall below the detection threshold. Additionally, the application of a probabilistic access rate to a discrete number of households represents a necessary methodological simplification that may not capture household-level connectivity dynamics.

WorldPop datasets are used for building count estimates and urban-rural delineation. While derived from high-resolution satellite imagery, the building count data carries potential for both false positives (classifying non-structures as buildings) and false negatives (missing small structures), particularly in complex rural terrains or dense informal settlements. A more significant uncertainty arises from using these counts to spatially disaggregate census households. Our method implicitly assumes a constant "households per building" ratio within a province, as it cannot distinguish between residential dwellings and non-residential structures (e.g., schools, shops, or agricultural storage). This may inadvertently over-allocate households and thus electricity demand to cells containing clusters of non-residential buildings, while under-estimating density in areas with multi-household dwellings that appear as single footprints. Nevertheless, in the absence of granular building classification data, a consistent residential allocation remains the most robust approximation available. Finally, the delineation between urban and rural areas relies on density thresholds that may not perfectly align with official administrative definitions, introducing uncertainties in the assignment of households counts and thus electricity consumption.

The Demographic and Health Survey (DHS) is the foundational dataset for our residential consumption profiles, providing a statistically representative sample of households. However, it relies on self-reported appliance ownership and does not include an exhaustive inventory of all possible household appliances. Furthermore, our derived electricity consumption values are based on standardised appliance ratings that may not perfectly reflect the specific age, efficiency, or usage patterns of equipment found in Zambian households. Finally, the sampling design based on household selection within clusters may fail to capture the full heterogeneity of niche communities or highly localised consumption behaviours.

The relative wealth index (*rwi*) is used as a spatial proxy for household wealth. While derived from a machine learning algorithm trained on multi-country DHS data, the *rwi* is generated at a global scale. Consequently, discrepancies arise when compared to the Zambian DHS data (as detailed in Section 2.1.2), and the necessary calibration introduces a degree of uncertainty. Additionally, because the *rwi* represents the average wealth across a 5.76 km² area, it inherently smooths over local, on-the-ground inequality. The accuracy of the index is also contingent on the quality of the underlying digital connectivity data. Finally, because the *rwi* is trained on a combined national wealth index rather than separate urban and rural indices, it fails to fully capture the structural differences between urban and rural wealth profiles.

For the mining consumption, the USGS dataset provides the most comprehensive public inventory of mining sites. However, its data on production capacity is not updated annually and contains some gaps. This necessitated significant preprocessing, including filling missing values and scaling capacities from 2017 to our 2019 baseline using national production totals, a source of uncertainty. Furthermore, assigning specific processing technologies based on keywords is a simplification that may not capture all equipment used at every facility. Proprietary datasets could offer a more precise picture of processing equipment and mining locations, for instance including artisanal and small-scale mining, but would go against the open-source principle of this methodology. However, the omission of the small-scale sites has probably a minimal impact on the results, as these operations are less likely to be connected to the main grid.

3.3.2. Limitations

To quantify the impact of these uncertainties and transparently acknowledge the model's limitations, several sensitivity analyses were conducted on some key parameters (see also Supplementary Material 5). The results demonstrate that while the absolute consumption total is sensitive to certain core assumptions, the primary spatial findings are highly robust. One of the parameters that has the most impact on the results is the calibration factor for electrification rate applied to the HREA data in urban areas. Removing it increases the total unscaled residential consumption estimate to almost 3,500 GWh and also leads to an unrealistically high urban electricity access rate of 96 %, validating the necessity of our correction. Another high impact parameter is the appliance consumption table. Adjusting the consumption values for the three most consuming appliances (refrigerators, air conditioners and "other electricity") by ± 20 % resulted in a corresponding ± 20 % change in total residential demand. Similarly, changing the elasticity of electricity use or the household nominal size linked to the appliances table also has a strong impact on the results. For instance, changing the household nominal size from 4 to 5 decrease the total to 2,562 GWh while changing the elasticity to 0.3 instead of 0.4 leads to a value of 2,728 GWh. The key finding from this analysis is that while the absolute national consumption total is moderately sensitive to these parameters, the core spatial patterns of demand—the heavy concentration in Lusaka and the Copperbelt and the stark urban-rural divide—remained highly robust across all sensitivity scenarios.

Beyond the sensitivity analysis tested above, the model has several structural limitations inherent to its bottom-up approach. First, the finite cell size inevitably hides any heterogeneity within a cell. Second, the model does not account for certain behaviours such as informal grid connections or electricity theft, which could lead to an underestimation of true consumption in some areas. Finally, the accuracy of the results is highly dependent on the quality of the multiple input datasets that have already been discussed in the above section.

Another conceptual limitation is that the model maps current consumption patterns based on existing appliance ownership as reported in the DHS. This approach does not fully capture latent demand—the unmet electricity demand due to various constraints. In areas with suppressed demand due to frequent outages or affordability constraints, our k-NN model likely underestimates the true consumption potential, even for connected households. However, this limitation also points to a powerful application and a key avenue for future research. The established relationship between wealth (*rwi*) and consumption in the model provides a framework to estimate this latent demand. By simulating full and reliable electricity access for all households within a cell, the methodology could be used to generate a "latent demand map." Such a map would be useful for infrastructure planning, by taking into account the demand that will emerge as access and incomes grow.

Reconciling the remaining gap between our bottom-up residential estimate (2,803 GWh) and the national statistics (4,023 GWh) represents another important avenue for future research. This work could involve improving remote sensing inputs with ground-truth data, refining

appliance consumption estimates with local surveys, and better understanding the scale of informal electricity use. Despite these limitations, our methodology provides a transparent and spatially explicit framework that significantly advances the understanding of electricity demand patterns in Zambia.

4. Conclusion

This study presents a new and open-source methodology for highly spatially disaggregated electricity consumption assessment in the residential, services and industry sectors of Zambia. By combining GIS datasets with national-level statistics, our bottom-up approach for the residential and industrial sectors, complemented by a top-down model for services, reveals the stark spatial inequality of electricity consumption. Consumption is heavily concentrated in urban and industrial hubs, particularly Lusaka and the Copperbelt, while vast rural areas remain low-consumption zones.

Our analysis also highlighted discrepancies between bottom-up estimates and aggregated national statistics. To reconcile these discrepancies and ensure alignment with the UN statistics, results can be scaled and other alternative allocation methods are detailed in Supplementary Material 4. These gaps underscore inherent uncertainties in the different data sources, for instance in the case of electricity access rates, the quality of the HREA dataset, or the appliance coverage in the DHS survey. By providing a transparent framework, our methodology not only generates a crucial dataset but also pinpoints conflicting data sources, whose collection and reporting could be improved. Designed for replicability, our open-source approach can be readily extended to other LMICs with available DHS survey data or similar surveys by substituting the country specific GIS input files. For data-scarce regions lacking recent surveys, the model can be adapted by deriving consumption parameters from DHS data of neighbouring or socio-economically similar countries. Future research can also focus on including additional sectors like agriculture, for which crops and irrigation can be used as a proxy for electricity consumption, as well as other energy-intensive industrial sectors. By leveraging the identified sector-specific drivers for demand, projections of electricity demand can also be undertaken based on drivers like new mining operations, production levels of key minerals (e.g., copper, cobalt) or demographic shifts.

In addition, the high spatial resolution of our electricity consumption estimates can contribute to inform more effective and equitable energy policy, directly supporting Sustainable Development Goal 7 (Affordable and Clean Energy) and 10 (Reduced Inequalities). Using this spatially resolved electricity consumption baseline, future demand can be projected via spatially differentiated drivers, such as applying specific urbanisation rates to hexagon cells, simulating wealth-driven appliance uptake via *rwi* growth, or modelling different production quantities or new mines as discrete industrial point-loads. The resulting spatially explicit demand projections can provide valuable inputs for power system models, informing grid planning and infrastructure investment decisions or decentralized generation planning. Incorporating appliance-specific demand profile, particularly for cooling, can also inform strategies to manage peak demand. These demand projections could also be adapted to provide an assessment of the latent electricity demand for households without access, based on the relationship between consumption and the *rwi* of the area described in our methodology. This facilitates targeted electrification planning in areas with high unmet need. Finally, by linking electricity consumption with *rwi* trends and other socioeconomic indicators (e.g., gender data), this research can also provide valuable insights into the socio-economic distributional impact of energy transitions, impacts on marginalised groups and energy poverty and thus contribute to more equitable energy access.

Data availability

The list of data used for the analysis is described in Supplementary

Material 1. They are all open-source and can directly be downloaded. The code to run the analysis is available at <https://github.com/EnergySystemsModellingLab/EDeMOS> and the results data on Zenodo at <https://doi.org/10.5281/zenodo.17910862>.

Declaration of generative AI and AI-assisted technologies in the writing process

Statement: During the preparation of this work the author A.M. used Gemini 2.5 Pro in order to improve readability and language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Ariane Millot: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Andelka Kerekes:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis. **Alexandros Korkovelos:** Writing – review & editing, Software, Methodology. **Martin Stringer:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Software. **Adam Hawkes:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Given his role as an Editor for Renewable and Sustainable Energy Transition, Adam Hawkes had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor.

If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.rset.2026.100138](https://doi.org/10.1016/j.rset.2026.100138).

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